

DUC AN NGUYEN, PhD

Computer Vision & Edge AI Engineer

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 France

 Talent – Researcher residence permit

SUMMARY

7+ years across research and industry, spanning computer vision, embedded systems, and applied AI. Skilled in the full computer vision lifecycle, from algorithm and model development to optimization and integration into real-time systems. Built multi-camera AI systems processing 24+ camera streams while reducing GPU resource requirements by 50%+ compared with independent vision pipelines. Hands-on deployment across NVIDIA Jetson, Raspberry Pi, Arm-based platforms and GPU infrastructure.

PROFESSIONAL EXPERIENCE

INDEPENDENT PROJECT

May 2026 - present
France

Computer Vision R&D | Infrared Object Detection

- Developed object-detection models for nighttime monitoring in low-contrast infrared imagery
- Analyzed detection failures across varying levels of contrast, noise, and image quality to guide model refinement

Technologies: Python, PyTorch, ONNX, TensorRT, Linux, Git, DVC

PRODUCTION ENGINEERING LABORATORY

Feb 2023 - Jan 2026
France

PhD Researcher | Applied AI & Explainable AI ([View publications ↗](#))

- Deployed an edge-cloud health monitoring system on **NVIDIA Jetson Orin Nano** using MQTT, Docker, PostgreSQL, and Grafana, and integrated it with a LoRA-based federated learning
- Designed a SHAP-based explainability framework for **multimodal deep learning** to diagnose prediction failures and reveal how different data types influence model decisions
- Developed an explainable autoencoder framework to address key challenges in industrial health monitoring, including **automatic sensor selection** and tracing the degradation source
- **First author of 7 publications:** 3 peer-reviewed journal articles, 3 international conference papers, and 1 book chapter

Technologies: Python, TensorFlow, Git, Jetson Orin Nano, Docker, PostgreSQL, MQTT, Grafana

NOVA INTELLIGENT TECHNOLOGY JSC

Oct 2021 - Jan 2023
Vietnam

AI Engineer | Computer Vision ([View Project ↗](#))

- Architected HydraNet, a multi-threaded shared-backbone **multi-camera vision architecture**, reducing GPU compute/VRAM requirements by 50%+ versus independent pipelines
- Integrated and optimized the pipeline with **Yolo-based object detection**, face recognition, fire detection, and SORT tracking under a 4 GB VRAM constraint
- Optimized the AI inference pipeline using **ONNX, TensorRT**, and other compute optimization techniques, achieving 24+ camera streams on one NVIDIA RTX A5000 at 5–10 FPS per stream
- **Data preparation**, CVAT labeling, and **training** for object detection and segmentation
- Developed lightweight **image processing algorithms** to improve inference accuracy and computational efficiency

Technologies: Python, PyTorch, OpenCV, ONNX, TensorRT, Linux, Git, Docker, DVC

PROFESSIONAL EXPERIENCE (CONTINUED)

HO CHI MINH
UNIVERSITY OF
TECHNOLOGY
Aug 2018 - Aug 2021
Vietnam

Research Engineer | Embedded Computer Vision

Project: Coffee Bean Sorting Machine - Jetson Nano / Raspberry Pi ([View Project ↗](#))

- Designed and implemented an **end-to-end real-time** coffee bean sorting system, from hardware setup and image acquisition to a multi-process, multi-threaded vision pipeline
- Developed **image enhancement, denoising**, and segmentation algorithms for robust object isolation under non-uniform lighting and background conditions
- Investigated a CIELAB color space-based classification model using the Pocket algorithm, achieving over 92% accuracy under illumination changes
- Optimized embedded vision pipelines for memory usage and execution speed, sustaining 20+ FPS on **NVIDIA Jetson Nano** under constrained compute and memory resources

Technologies: C/C++, NVIDIA Jetson Nano, Raspberry Pi, OpenCV, MATLAB, Memory optimization

Project: Embedded Fingerprint Recognition System - STM32 (C)

- Implemented fingerprint enhancement, segmentation, thinning, and feature-extraction algorithms to extract reliable ridge features for matching under high-noise conditions
- Improved and adapted the Lin Hong matching algorithm for efficient fingerprint comparison
- Optimized memory and computational efficiency to meet STM32 deployment constraints

Technologies: C/C++, STM32, MATLAB, Memory optimization

Other projects:

- Developed a real-time object detection system using MobileNet and OpenCV on **Raspberry Pi**
- Controlled storage temperature and logged data to an SD card with the **Tiva C kit**
- Designed a MIPS processor using Verilog

SKILLS

Programming & frameworks: Python, C/C++, MATLAB, PyTorch, TensorFlow, OpenCV

Edge deployment: NVIDIA Jetson, Raspberry Pi, STM32, TensorRT, TFLite, ONNX, Inference and Memory Optimization

Other tools: Linux, Git, DVC, Docker, PostgreSQL, MQTT, Grafana

LANGUAGES

English (B2-C1)

Vietnamese (native)

French (A2-B1)

EDUCATION

PhD, Toulouse INP
Applied AI & Explainable AI - Industrial & Computer Engineering

Feb 2023 - Jan 2026
France

M.Sc., Ho Chi Minh City University of Technology
Embedded Computer Vision - Electronic Engineering

Sep 2019 - Sep 2021
Vietnam

B.S., Ho Chi Minh City University of Technology
Embedded Systems - Electronics and Telecommunications Engineering

Sep 2014 - Apr 2019
Vietnam